L76K GPS Module

From Waveshare Wiki
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Overview

Introduction

L76K is a multi-satellite system (GPS, BeiDou, GLONASS, QZSS), multi-system joint positioning and single system independent positioning, supports AGNSS function, built-in low noise amplifier and surface acoustic filter, can provide users with fast and accurate, GNSS module with high-performance positioning

L76K/LC76G/L76B GPS Module



(https://www.waveshare.com/l76k-gps-module.htm)

GNSS/UART

experience.

This module supports Raspberry Pi, Arduino, Raspberry Pi Pico and ESP32 control demo.

Parameters

Model	LC76G(AB)	L76K	L76B
Compatible			
Picture	 (/wiki/File:LC76G_p.png)	 (/wiki/File:L76K_p.png)	 (/wiki/File:L76B_p.png)
GNSS	GPS/GLONASS/Galileo/BDS/QZSS	GPS/GLONASS/BDS/QZSS	GPS/BDS/QZSS
Form Factor	LCC	LCC	LCC
Dimensions	9.7×10.1×2.4	9.7×10.1×2.4	9.7×10.1×2.4
Weight (g)	0.6	0.4	0.5
Normal Operating Temperature	-40℃ ~+85℃	-40℃ ~+85℃	-40℃ ~+85℃
Storage Temperature	-40℃ ~+90℃	-40℃ ~+90℃	-40℃ ~+90℃
Main Feature			
Function	Standard	Standard	Example
Chip	AG3352Q	AT6558R	MT333
L1 Receiver (Channel)	47 tracking ch	32 tracking ch/72 acquisition ch	22 tracking ch/66 acquisition ch
L1 Receiver SBAS	Support	N/A	Support
A-GNSS	Support	Support	Support
Sensitivity (Auto-aquisition)	-147dBm	-148dBm	-148dBm
Sensitivity (re-acquisition)	-159dBm	-160dBm	-160dBm
Sensitivity (tracking)	-166dBm	-162dBm	-163dBm
TTFF (Time To First Fix) Cold Starts	26s, Autonomous 12s, with EASY™ 5s,with EPO™	30s, Autonomous 5.5s, With AGNSS	35s, Autonomous 15s, with EASY™
TTFF (Time To First Fix) Warm Starts	20s, Autonomous 2s, with EASY™	/	30s, Autonomous 5s, with EASY™

TTFF (Time To First Fix) Hot Starts	1s	2s	1s
Position Accuracy	1.5m CEP	2.0m CEP	2.5m CEP
Velocity Accuracy	0.1m/s	0.1m/s	0.1m/s
Acceleration Accuracy (Max)	0.1m/s²	0.1m/s²	0.1m/s²
Timing Accuracy	100ns	30ns	10ns
Update Rate (Max)	10Hz	5Hz	10Hz
Default Baudrate	115200bps	9600bps	9600bps
Geo-fence	⚪	/	⚪
Interference Detection	⚪	/	⚪
Anti-jamming	⚪	/	⚪
LNA	⚪	⚪	⚪
Electrical Parameters			
Power Supply	2.55 to 3.6V	2.7V to 3.4V	2.8V to 4.3V
Power Consumption (Acquisition)	36mA(G3B)	29mA (GPS+BDS)	29mA (GPS+BDS)
Power Consumption (Tracking)	36mA(G3B)	29mA (GPS+BDS)	18mA (GPS+BDS)
Power Consumption (RTC backup/low power consumption	13μA	8μA	7μA
Interface			
UART	⚪	⚪	⚪
IIC (NMEA)	⚪	/	⚪
Reset	⚪	⚪	⚪
Timing Pulse	⚪	⚪	⚪
Antenna			
Short circuit and open circuit detection	⚪	⚪	⚪
Antenna Type	Active, Passive	Active, Passive	Active, Passive
Antenna Power Supply	External or Internal	External or Internal	External or Internal

Interface

Interface	Description
VCC	Power supply (2.7V~5V)
GND	Ground
TX	UART data output
RX	UART command input
PPS	Status indication (output second pulse if the positioning is successful)

NMEA0183

- The host control output NMEA0183 information from the serial port and analyzes NMEA 0183 sentence to output the information.
- NMEA 0183 is a standard format developed by the National Marine Electronics Association for marine electronic equipment. It has now become a unified RTCM (Radio Technical Commission for Maritime Services) standard protocol for GPS navigation equipment.
- NMEA 0183 includes 7 protocol frames including $GPZDA, $GPRMC, $GPVTG, $GPGNS, $GPGGA, $GPGSA, $GPGSV*3, $GPGLL, $GPGST, among which the first two characters followed by $ represent the country or region. For example, GPGGA stands for American GPS, BDGGA stands for China Beidou, GLGGA stands for Russia GLONASS, GAGGA stands for EU Galileo, and GNGGA stands for multi-satellite joint positioning.
- Take $GPRMC as an example to briefly describe the information represented by each part of the protocol frame. Please refer to the NMEA 0183 manual (https://files.waveshare.com/upload/5/57/NMEA0183.pdf) for the other 6 protocol frames.

```
Recommended Minimum Specific GPS/TRANSIT Data (RMC) recommended positioning infor
mation
$GPRMC,<1>,<2>,<3>,<4>,<5>,<6>,<7>,<8>,<9>,<10>,<11>,<12> *hh<CR><LF>
$GNRMC,010555.000,A,2232.4682,N,11404.6748,E,0.00,125.29,230822,,,D*71
<1> UTC time, hhmmss.sss (hour, minute, second) format
<2> Positioning state, A=valid positioning, V=invalid positioning
<3> Latitude ddmm.mmmm (degrees and minutes) format (the front 0 will also be tra
nsmitted)
<4> Latitude hemisphere N (Northern Hemisphere) or S (Southern Hemisphere)
<5> Longitude dddmm.mmmm (degrees and minutes) format (the preceding 0 will also
be transmitted)
<6> Longitude hemisphere E (East longitude) or W (West longitude)
<7> Ground speed (000.0~999.9 knots, the preceding 0 will also be transmitted)
<8> Ground heading (000.0~359.9 degrees, with true north as the reference, the pr
eceding 0 will also be transmitted)
<9> UTC date, ddmmyy (day month year) format
<10> Magnetic declination (000.0~180.0 degrees, the previous 0 will also be trans
mitted)
<11> Magnetic declination direction, E (East) or W (West)
<12> Mode indication (only NMEA0183 version 3.00 output, A=autonomous positionin
g, D=difference, E=estimation, N=data invalid)
*hh: The last check code *hh is the data used for parity. In normal use, it is no
t necessary, but when there is strong electromagnetic interference in the surroun
ding environment, it is recommended to use. hh represents the bitwise XOR value o
f all characters of "$" and "*" (excluding these two characters). Individual manu
facturers define their own statement format starting with "$P", followed by a 3-c
haracter manufacturer ID number, followed by a custom data body.
```

PCAS Sentence

PCAS sentences are NMEA-specific sentences defined by chip suppliers.

Statement	Description	Example	Function
$PCAS01,<CMD>*<Checksum>	configure NMEA serial port baud rate	$PCAS01,1*1D $PCAS01,5*19	set serial port baud rate to 9600 <CMD>The following baud rates are supported: 0 = 4800 1 = 9600 2 = 19200 3 = 38400 4 = 57600 5 = 115200
$PCAS02,<Interval>*<Checksum>	set positioning frequency	$PCAS02,1000*2E	set positioning frequency to 1Hz <Interval>Positioning interval: 1000: set the positioning frequency to 1 Hz 500: Set the positioning frequency to 2 Hz 200: Set the positioning frequency to 5 Hz
$PCAS03,<nGGA>,<nGLL>,<nGSA>,<nGSV>,<nRMC>,<nVTG>,<nZDA>,<nANT>,<Res>,<Res>,<Res>,<Res>,<Res>,<Res>*<Checksum>	Configure NMEA sentence output type and output frequency	$PCAS03,1,1,1,1,1,1,1,1,0,0,,,0,0*02	Enable GGA, GLL, GSA, GSV, RMC, VTG, ZDA, ANT sentence output in NMEA output (see L76K&L26K GNSS protocol specification: 2.3 PCAS sentence (special sentence))
$PCAS04,<Mode>*<Checksum>	Configure Galaxy	$PCAS04,3*1A	Configure Galaxy as GPS+BeiDou <Mode>GNSS galaxy configuration: 1 = GPS 2 = BeiDou 3 = GPS + BeiDou (default) 4 = GLONASS 5 = GPS + GLONASS 6 = BeiDou + GLONASS 7 = GPS + BeiDou + GLONASS (QZSS is enabled by default and does not support configuration.)

$PCAS10,<Flag>*<Checksum>	Restart module	$PCAS10,0*1C	Hot start <Flag>Restart mode: 0 = warm start 1 = warm start 2 = cold start 3 = Cold boot and factory reset

Note: After changing the configuration content, you need to modify the checksum <Checksum> accordingly. For more details, you can refer to Quectel website (https://www.quectel.com/).

Working in Windows

- Connect the GPS antenna and the USB to the UART module (not included), and then connect to the computer. as the picture shows. After connecting, the power indicator light (PWR) is always on.



(/wiki/File:L76K_GPS_Module_win.png)

Note: The red cable is not VCC, and the black cable is not GND.

Module	USB To UART
VCC	5V/3.3V
GND	GND
RX	TXD
TX	RXD
PPS	N/A

- Open the computer serial port debugging assistant, set the corresponding serial port number (com5 here), baud rate 9600, 8 data bits, 1 stop bit, no parity bit, and no flow control.

Sscom can be downloaded Here (https://files.waveshare.com/upload/5/5f/Sscom.7z).



(/wiki/File:L76K_GPS_Module_win02.png)

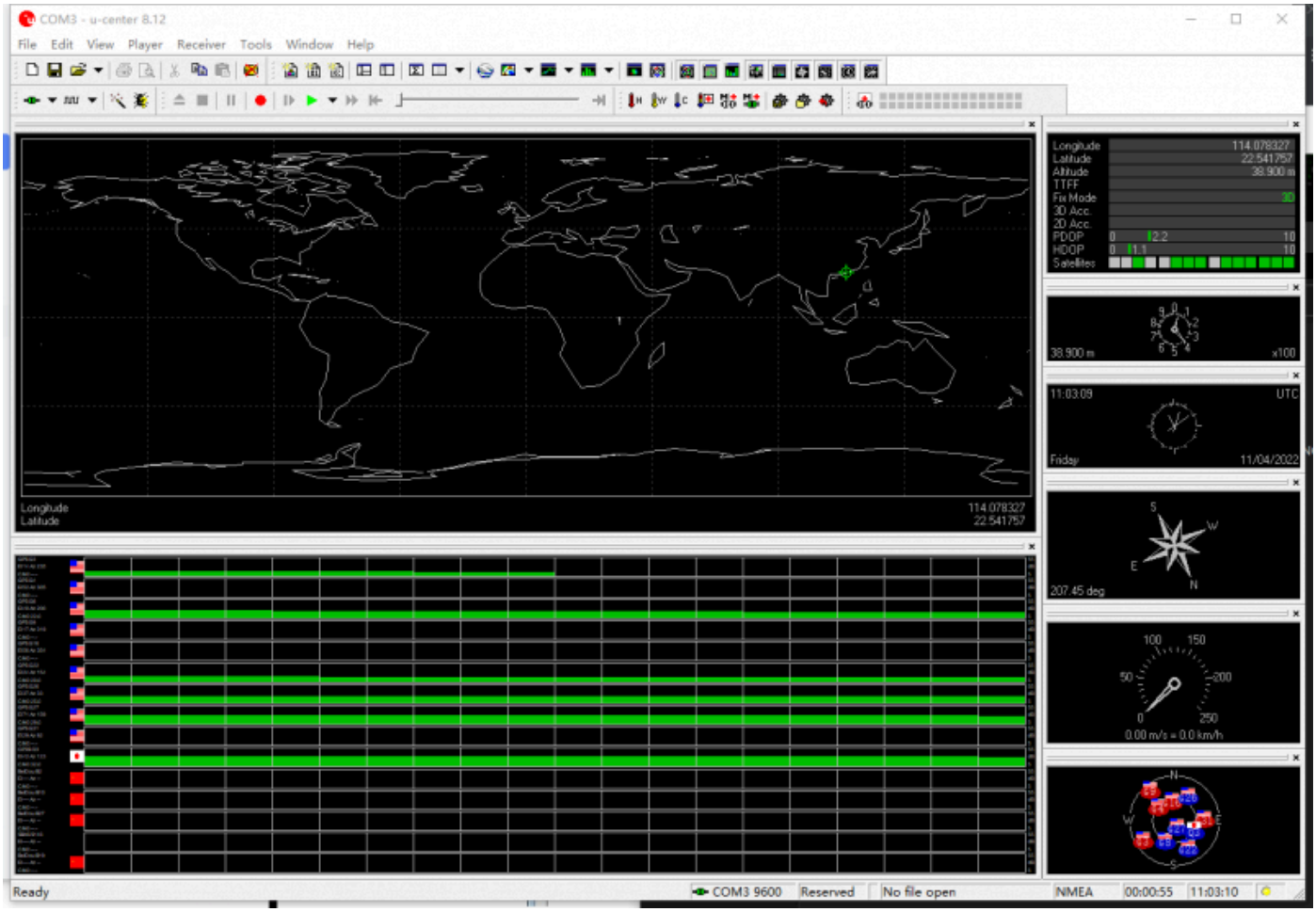
Note:

1. Due to the instability of GPS indoor search, please put the module or antenna next to the balcony or window, or conduct experiments directly outdoors.

2. The first positioning of the module (cold start), under normal circumstances (outdoors, good weather, no large building block), takes 35 seconds to successfully locate, please wait patiently. If the weather conditions are bad, it may take longer to locate, or even fail to locate.

Download U-centersetup v8.12.zip (https://files.waveshare.com/upload/a/a8/U-centersetup_v8.12.zip), unzip it, and open it for installation. Run the program, click the Receiver menu, select Port, and set the actual serial port number, Baudrate: 9600. Click (connect button) to connect to the GPS

module, and the u-center will display various information, as shown below:



(/wiki/File:L76K_GPS_Module_win03.png)

Working With Raspberry Pi

Provide C and Python demos for control in Raspberry Pi.

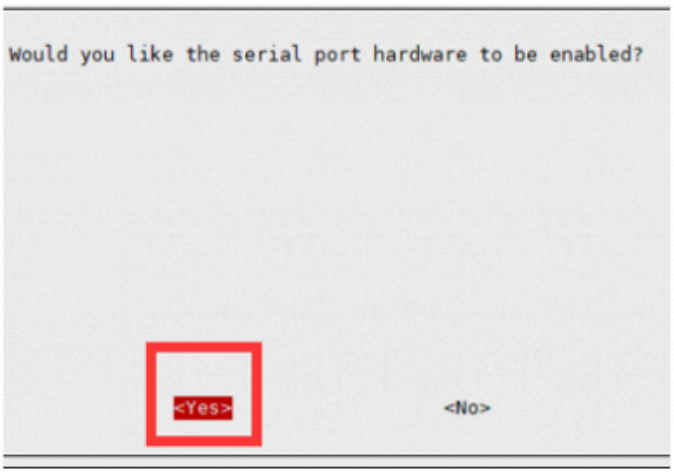
Enable UART Port (Skip if enabled)

Enable the Raspberry Pi terminal and input the following commands to enter the interface.

```
sudo raspi-config
```

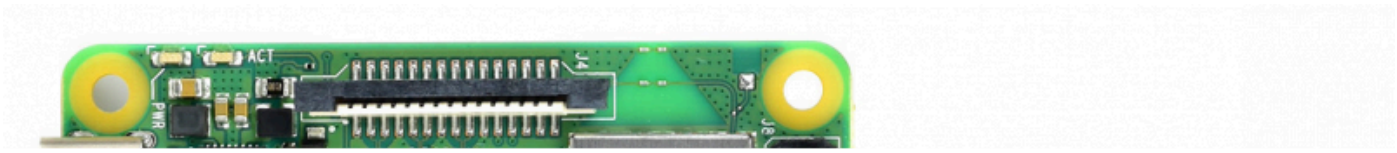
Choose Interfacing Options -> Serial, close the shell visit, and enable the serial port of the hardware:





(/wiki/File:L76K_GPS_Module_RPI.png)

Hardware Connection



(/wiki/File:L76K_GPS_Module_RPI02.png)

Module	Raspberry Pi
VCC	5V
GND	GND
TX	P15
RX	P14
PPS	NC

Minicom Debug

Download and install minicom tool:

```
sudo apt-get install minicom
```

The corresponding serial port number for the position hardware serial port:

```
ls -l /dev/serial*
```

If the system has used the serial port and has been modified, there are two scenarios:

The default state after enabling the UART function:

```
pi@raspberrypi:~ $ ls -la /dev/serial*
lrwxrwxrwx 1 root root 5 Mar 17 13:38 /dev/serial0 -> ttyS0
lrwxrwxrwx 1 root root 7 Mar 17 13:38 /dev/serial1 -> ttyAMA0
pi@raspberrypi:~ $
```

(/wiki/File:Raspberry_Pi_Documentation-GPIO31.png)

The state after modifying the UART config.

```
pi@raspberrypi:~ $ ls -la /dev/serial*
lrwxrwxrwx 1 root root 7 Mar 17 13:44 /dev/serial0 -> ttyAMA0
lrwxrwxrwx 1 root root 5 Mar 17 13:44 /dev/serial1 -> ttyS0
pi@raspberrypi:~ $
```

(/wiki/File:Raspberry_Pi_Documentation-GPIO32.png)

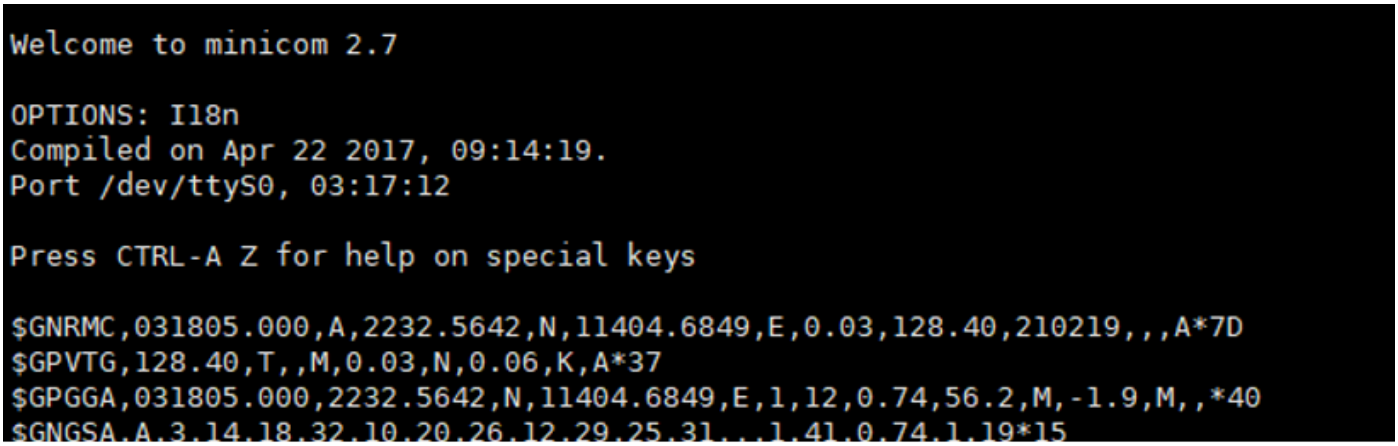
Need to find the serial port number corresponding to serial0 (that is, the serial port on GPIO), as shown in Figure 1, the serial port number is ttyS0, and Figure 2 is ttyAMA0; use minicom to open the serial port:

```
sudo minicom -D /dev/ttyS0 -b 9600
```

-D stands for port. If the serial port number in the previous step is ttyAMA0, then it is:

```
sudo minicom -D /dev/ttyAMA0 -b 9600
```

The default baud rate of minicom is 115200, if you need to set the baud rate to 9600, add the parameter -b 9600.



(/wiki/File:L76X_GPS_HAT_X3PI02.png)

Exit: Ctrl+A then press X alone, YES to enter.

Download and Run the Test Demo

Download Demo

```
cd
sudo apt-get install unzip -y
sudo wget https://files.waveshare.com/upload/0/05/L76K_GPS_Module_code.zip
sudo unzip L76K_GPS_Module_code.zip -d ./L76K_GPS_Module_code
sudo chmod 777 -R ./L76K_GPS_Module_code
cd ./L76K_GPS_Module_code
```

Or click to download the sample demo (https://files.waveshare.com/upload/0/05/L76K_GPS_Module_code.zip), unzip it, and put it into the Raspberry Pi.

Environment Config

Install the related function library of the demo.

BCM2835

```
#Open the Raspberry Pi terminal and run the following command
wget http://www.airspayce.com/mikem/bcm2835/bcm2835-1.71.tar.gz
tar zxvf bcm2835-1.71.tar.gz
cd bcm2835-1.71/
sudo ./configure && sudo make && sudo make check && sudo make install
# For more information, please refer to the official website: http://www.airspayce.com/mikem/bcm2835/
```

WiringPi

Open the Raspberry Pi terminal and run the following commands:

```
cd
sudo apt-get install wiringpi
#For Raspberry Pi systems after May 2019 (earlier ones do not need to be executed), an upgrade may be required:
wget https://project-downloads.drogon.net/wiringpi-latest.deb
sudo dpkg -i wiringpi-latest.deb
gpio -v
# Run gpio -v and version 2.52 will appear, if there is no installation error
```

```
The #Bullseye branch system uses the following commands:
git clone https://github.com/WiringPi/WiringPi
cd WiringPi
./build
gpio -v
# Run gpio -v and version 2.70 will appear if there is no installation error
```

Python

The new version of the Raspberry Pi system needs to install the GPS data analysis library to run the python3 demo.

```
cd L76K_GPS_Module_code/RaspberryPi/python3/micropyGPS-master
sudo python setup.py install
#Wait for the library installation to complete
```

If the old system only supports Python2, the function library is required.

```
sudo apt-get update
sudo pip install RPi.GPIO
sudo apt-get install python-serial
```

C Demo

```
cd C
make clean
make
sudo ./main
```

Baidu Coordinates is the converted Baidu map coordinates, copy the coordinates to http://www.gpsspg.com/maps.htm (http://www.gpsspg.com/maps.htm), select Baidu map on the left, and you can see your location (if you go directly to "Baidu Pickup Coordinate System" you need

Swap the positions of the two latitude and longitude numbers).



(/wiki/File:L76K_GPS_Module07.png)

Python 3 Demo

```
cd python
sudo python main.py
```

Expected Result

It takes about 35 seconds for the module to be positioned for the first time.

The front is the raw data output by the module.

"Time" is the time output by L76X GPS Module.

Latitude and longitude are the output latitude and longitude and latitude and longitude direction.

Baidu Coordinates is the converted Baidu nao coordinates, copy the coordinates to http://www.gpsspg.com/maps.htm (http://www.gpsspg.com/maps.htm), select Baidu map on the left, and you can see your location (if you go directly to "Baidu Pickup Coordinate System" you need

to "Swap" the positions of the two latitude and longitude numbers).



(/wiki/File:L76K_GPS_Module08.png)

Python 2 Demo

```
cd python
sudo python main.py
```

Expected Result

It takes 35 seconds for the module to be positioned for the first time.

The front is the raw data output by the module.

Time is the time output by L76X GPS Module.

Latitude and longitude are the output latitude and longitude and the directions of latitude and longitude.

Baidu Coordinates is the converted Baidu map coordinates, copy the coordinates to http://www.gpsspg.com/maps.htm (http://www.gpsspg.com/maps.htm), select Baidu map on the left, and you can see your location (if you go directly to "Baidu Pickup Coordinate System" you need

Swap the positions of the two latitude and longitude numbers).

```
,1.10,0.78,0.78*1E
$GPGSV,4,1,13,10,58,179,25,32,56,010,41,42,50,127,,31,49,292,40*79
$GPGSV,4,2,13,14,46,343,36,25,46,051,31,20,30,165,32,26,26,205,30*7E
$GPGSV,4,3,13,12,13,038,27,29,12,111,23,22,08,320,41,18,07,276,*7A
$GPGSV,4,4,13,193,,,*40
$GLGSV,2,1,06,65,72,092,,76,67,303,,75,39,017,27,66,33,338,33*6E
$GLGSV,2,2,06,72,29,134,,77,21,231,*68
$GNGLL,2232.6312,N,11404.6593,E,032729.171,A,A*43
$GPZDA,032729.571,21,02,2019,,*53
$GNRMC,032729.571,A,2232.6331,N,11404.6643,E,0.58,297.14,210219,,,A*72
$GPVTG,297.14,T,,M,0.58,N,1.08,K,A*30
$GPGGA,032729.571,2232.6331,N,11404.6643,E,1,12,0.78,131.9,M,-1.9,M,,*70
$GNGSA,A,3,10,32,31,14,25,20,26,12,22,29,,,1.10,0.78,0.78*12
$GNGSA,A,3,66,75,,,,,,,,,,,1.10,0.78,0.78*1E
$GPGSV,4,1,13,10,58,179,23,32,56,010,41,42,50,127,,31,49,292,38

Time: 11:27:29
Latitude and longitude: 22.326331  N  114.046643  E
Baudu Coordinates   22.547066,114.089375
```

(/wiki/File:L76K_GPS_Module09.png)

Working With Arduino

This demo has been verified on Arduino MEGA2560.

Click to download the sample demo (https://files.waveshare.com/upload/9/94/L76K_GPS_Module_Arduino.zip) or download it in the #Resource.

Hardware Connection

Module	Arduino
VCC	5V
GND	GND
TX	RX2
RX	TX2
PPS	NC

Demo Usage

Choose the Arduino mega2560 development board and the corresponding port, and upload the demo after verification.

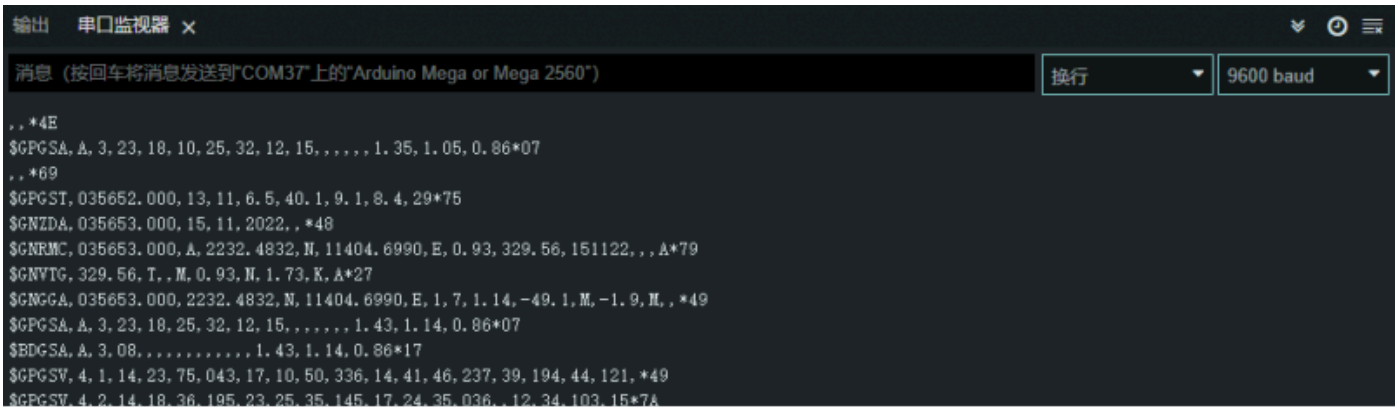
The first position for the module will take 35 seconds.

Open SSCOM and set the baud rate as 9600.

The original data is output by the module at first.

Time is the time of L76X GPS Module output.

Lat and Lon are the output latitude and longitude and latitude and longitude directions. B_Lat and B_Lon are the converted Baidu map coordinates, copy the coordinates to http://api.map.baidu.com/lbsapi/getpoint/ (http://api.map.baidu.com/lbsapi/getpoint/), and you can see your location (the coordinate format is longitude Lat first, latitude Lon after, separated by commas in the middle).



(/wiki/File:L76K_GPS_Module_arduino.png)

The Arduino uno demo is also included in the file, but the data accuracy of this platform is insufficient, and only the time in the satellite data is displayed. If you want other information, you can modify the program yourself.

Working With Raspberry Pi Pico

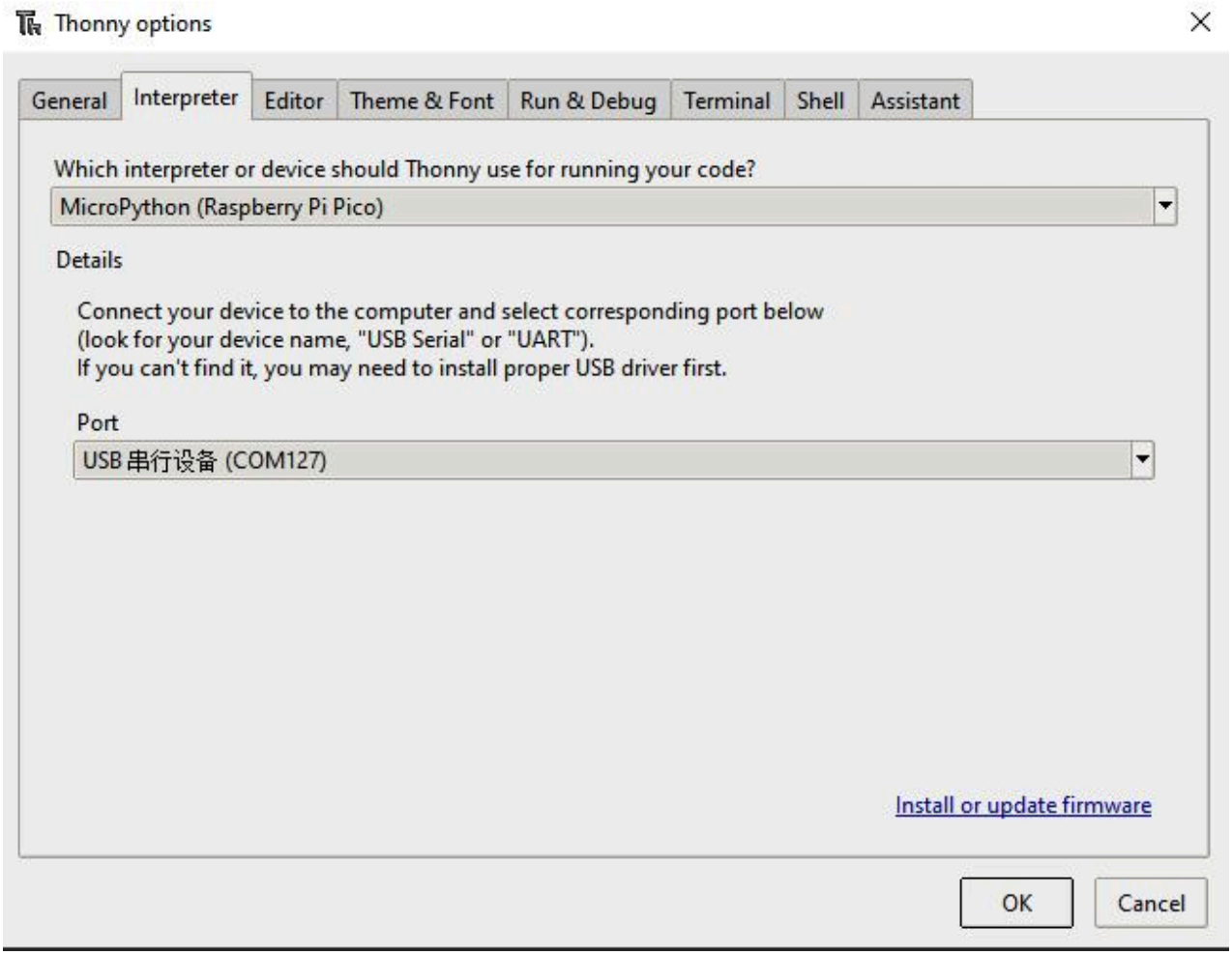
Hardware Connection

Module	Raspberry Pi Pico
VCC	5V
GND	GND
TX	GP1
RX	GP0
PPS	NC

Environment Building

We test the demo with Thonny (https://thonny.org/) in this tutorial, click to download the related IDE and open Thonny.

- Please refer to the official document (https://www.raspberrypi.com/documentation/microcontrollers/micropython.html) to set up the Python environment. Choose a Raspberry Pi Pico device in Tools->Options->Interprete. As shown below:

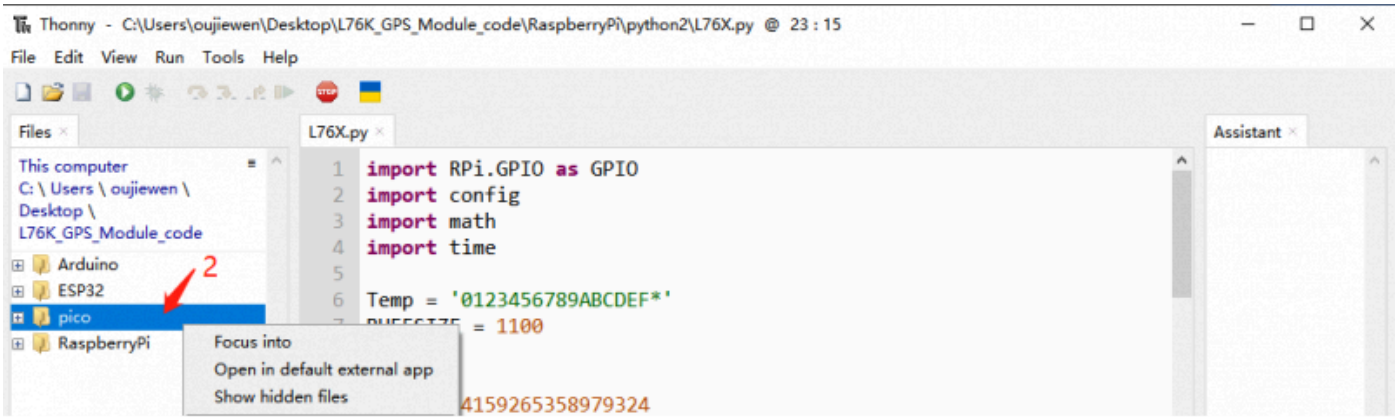


(/wiki/File:Pico-GPS-002.jpg)

Download Demo

1. Click to download the sample demo (https://files.waveshare.com/upload/0/05/L76K_GPS_Module_code.zip).
2. Unzip the sample demo.
3. Open Thonny, first confirm that you have connected to Pico, then open the decompressed demo

path in the upper left corner, right-click on the Pico folder, and select upload, as shown in the figure:



(/wiki/File:L76K_GPS_Module_Pico.png)

How To Use the Demo

1. Open Thonny IDE, choose the Pico directory, double-click the coordinate_convert.py file, and then run the demo, as shown below:

(/wiki/File:L76K_GPS_Module_Pico2.png)

2. Under sunny conditions, the L76K will obtain the positioning information about 30 seconds after it is powered on, as shown in the following figure when running the demo.

(/wiki/File:L76K_GPS_Module_Pico3.png)

3. The user can copy the positioning information displayed in the terminal in Thonny, and use Google Map (https://www.google.com/maps/@22.38131,114.168639,11z) and Baidu Map (https://api.map.baidu.com/lbsapi/getpoint/index.html) to mark the coordinates.

ESP32

Hardware Connection

(/wiki/File:L76K_GPS_Module_ESP32201.png)

Module	ESP32
VCC	5V
GND	GND
TX	P16
RX	P17
PPS	NC

Environment Building

Install ESP32 plug-in On Arduino IDE

1. Open the Arduino IDE, click on the file in the upper left corner, and select Preferences.

(/wiki/File:L76K_GPS_Module_ESP32202.png)

2. Add the following link in the additional development board manager URL, then click OK.

```
https://dl.espressif.com/dl/package_esp32_index.json (https://dl.espressif.com/dl/package_esp32_index.json)
```

(/wiki/File:L76K_GPS_Module_ESP32203.png)

Note: If you already have the ESP8266 board URL, you can separate the URLs with commas like this:

```
https://dl.espressif.com/dl/package_esp32_index.json, http://arduino.esp8266.com/stable/package_esp8266com_index.json (https://dl.espressif.com/dl/package_esp32_index.json, http://arduino.esp8266.com/stable/package_esp8266com_index.json)
```

3. Download the package (https://files.waveshare.com/upload/7/70/Packages.7z) and copy the packages file to the following path:

```
C:\Users\xutong\AppData\Local\Arduino15
```

(/wiki/File:L76K_GPS_Module_ESP32204.jpg)

Note: Replace the username: xutong with your own username.

How To Use the Demo

- Download the demo and unzip it, use the configured Arduino IDE to open the L76X.ino file in the ESP32 directory, and wait for all the files to be loaded.
- Use the data cable to connect the esp32 development board to the computer, and select the correct development board model and port number.
- Here we take the ESP32-S chip development board as an example: Click in turn: Tools->Development Board->esp32->ESP32 Dev Module, and select the corresponding port number.
- Click "Verify" in the upper left corner, wait for the verification to complete without error, and then click "Upload".
- After the upload is complete, the demo runs automatically. Click: Tools->Serial Monitor in turn, and you can observe the running effect of the demo in the serial monitor window:

(/wiki/File:L76K_GPS_Module_ESP32205.png)

It takes 35 seconds for the module to be positioned for the first time. (It may be longer if the signal condition is not good). The front is the raw data output by the module. Time is the time output by L76X GPS Module. "Lat" and "Lon" are the output latitude and longitude and the directions of latitude and longitude.

B_Lat and B_Lon are the converted Baidu map coordinates, copy the coordinates to

http://api.map.baidu.com/lbsapi/getpoint/ (http://api.map.baidu.com/lbsapi/getpoint/), and you can see your location (the coordinate format is longitude Lat first, latitude Lon after, separated by commas in the middle).

Jetson Nano

This example has been verified on the JETSON-NANO-DEV-KIT development board that programs the Ubuntu system.

Hardware Connection

(/wiki/File:L76K_GPS_Module_Jetson_Nano01.png)

Module	JETSON-NANO-DEV-KIT
VCC	5V
GND	GND
TX	10
RX	8
PPS	NC

Environment Debug

Install Pip Tool

Install Python tool pip with the apt-get tool.

```
sudo apt-get install python3-pip #python3
# After that, you need to enter the user password to verify
```

```
sudo apt-get install python2-pip #python2
# After that, you need to enter the user password to verify
```

Waiting for the installation.

Install Python Library

```
Libraries required by the #python3 program:
sudo pip3 install pyserial #serial port control library
sudo pip3 install settools #install tools
cd jetson/python3/micropyGPS-master
sudo python3 setup.py install #data parsing library
```

```
Libraries required by the #python2 program:
sudo pip install pyserial #serial port control library
```

How To Use the Demo

Enter the directory and run main.py.

```
cd jetson/python3
sudo python3 main.py
```

(/wiki/File:L76K_GPS_Module_Jetson03.png)

It takes 35 seconds for the module to be positioned for the first time.

The front is the raw data output by the module.

Time is the time output by L76X GPS Module.

Latitude and longitude are the output latitude and longitude and the direction of latitude and longitude.

Baidu Coordinates is the converted Baidu map coordinates, copy the coordinates to

http://www.gpsspg.com/maps.htm (http://www.gpsspg.com/maps.htm), select Baidu map on the left, and you can see your location (if you go directly to "Baidu Pickup Coordinate System" you need Swap the positions of the two latitude and longitude numbers).

Resource

Document

- L76X GPS Module Schematic (https://files.waveshare.com/upload/d/df/L76X_GPS_Module_SchDoc.pdf)
- 3D Drawing (https://files.waveshare.com/upload/f/f9/L76x_gps_module.zip)

Demo

- L76K GPS Module code (https://files.waveshare.com/upload/0/05/L76K_GPS_Module_code.zip)

Software

- Sscom (https://files.waveshare.com/upload/5/5f/Sscom.7z)
- U-centersetup v8.12 (https://files.waveshare.com/upload/a/a8/U-centersetup_v8.12.zip)
- TCP232 (https://files.waveshare.com/upload/8/88/TCP232.7z)
- Chiness_Unicode (https://files.waveshare.com/upload/c/c1/Chiness_Unicode.zip)
- Serial Bluetooth Terminal (https://files.waveshare.com/upload/5/56/Serial_Bluetooth_Terminal_v1.zip)
- Thonny (https://thonny.org/)

Datasheet

- For more resources, click here (https://www.quectel.com/cn/product/gnss-l76k)

FAQ

Question: Why the location of the running program is "00, 00"?

Answer:
Please place the antenna in an outdoor open area, with the word on the antenna facing down, and you can receive valid positioning data after 45 seconds of power on.

Question: How can I change the baud rate to 115200?

Answer:

Send the following commands to chafe the baud rate as 115200:

```
$PMTK251,115200*1F<CR><LF>
```

Send the following commands to restore the baud rate as 9600:

```
$PMTK251,9600*17<CR><LF>
```

Question: Why does the program I run once reports an error or ends at the beginning?

Answer:

(/wiki/File:L76X_GPS_HAT_FAQ01.png)

Make sure that the first step on the side is closed because this will also output the login shell information of the Raspberry Pi to the serial port, resulting in a serial port conflict.

(/wiki/File:L76X_GPS_HAT_FAQ003.png)

Question: Why is the program and minicom debugging also garbled, but using USB to serial port is normal?

Answer:

(/wiki/File:L76X_GPS_HAT_FAQ02.png)

If the above phenomenon occurs during use, please check the Raspberry Pi config.txt configuration, sudo nano /boot/config.txt, please block the following statement, this statement is that the temperature sensor occupies the serial communication line;

(/wiki/File:L76X_GPS_HAT_FAQ03.jpg)

Question: What is the function of STANDBY?

Answer:
It is in standby mode, please set it to OFF, otherwise it will stand by when it is turned on, and there is no GPS output.

Question: How can I change the refreshing rate of the GPS python program?

Answer:
In main.py in the python program, the 12th line is to modify the update rate. The default is 400ms. You can find the corresponding macro definition in L76X.py and replace it.

(/wiki/File:L76X_GPS_HAT_FAQ004.png)

(/wiki/File:L76X_GPS_HAT_FAQ005.png)

Question: About one minute after the power is turned on, the power indicator (PWR) is on, the TXD is not flashing, and the serial port has no data output?

Answer:
Check whether the STANDBY switch is in the OFF position, and then press the FORCE_ON button for about 1 second to observe whether the serial port has data output. If there is still no data output, check if the TXD and RXD pins are connected correctly.

Question: Why is the location not accurate?

Answer:
The positioning accuracy is related to the environment in which it is located. First, it is rainy due to weather and there is a lot of moisture in the air, which affects the transmission of signals. This is also the reason why the mobile phone signal is weaker in summer. It is rainy and humid in summer, coupled with high-temperature evaporation, which increases the moisture in the air, thus affecting the transmission of satellite signals. Due to the factor of two high-rise buildings, it is easy to cause poor satellite signals or signal offset problems next to some high-rise buildings. In low-rise buildings or underground buildings, such as underground parking lots, underground shopping malls, subways, tunnels, etc., due to the occlusion of walls, indoor signal attenuation is very large, which forms a weak signal coverage, so the positioning is inaccurate, the error is large, etc., and it is also easy to cause the positioning to be inaccurate under the obstruction of some mountains. The third is the number of satellites. The number of satellites placed over rural and remote areas is small, resulting in large position deviations. Four satellite signals will be affected by factors such as atmospheric ionosphere, ground buildings, forests, water surfaces, etc., resulting in deviations in calculations.

Question: Why am I still not receiving data when I'm outdoors?

Answer:
Please check whether the antenna is connected well, and the side with words should face down, and the receiving side of the antenna is on the side without words.

Question: Why is it inconsistent with the location of the mobile phone?

Answer:
This module positioning only relies on satellite positioning alone, and mobile phone positioning not only relies on satellite positioning but also relies on the combination of various positioning systems such as AGPS (Assisted Global Satellite Positioning System), and LBS base station positioning, Wifi positioning, Bluetooth positioning, etc. The positioning has a faster positioning speed, and the accuracy is different in different environments, and the multiple satellite systems supported by the GPS signal of the mobile phone are inconsistent with the multiple satellite systems supported by the module. Therefore, there is a certain difference between the data of the module and the mobile phone positioning.

Question: Why the module baud rate have not been changed after sending the change command?

Answer:
First, ensure that the baud rate of the command sent is the same as the baud rate output by the serial port. When the module searches for a large number of satellites, the baud rate setting from high to low may not be able to be modified, because the baud rate is too low. To send all data, consider using the SET_NMEA_OUTPUT command to reduce the output data field, and then modify the baud rate.

Question: What are the functions of jumper caps A, B, and C?

Answer:
The function of the jumper cap is to switch the serial port device. When the jumper cap is connected to A, the L76B will be connected to the USB interface; the jumper cap is connected to B, and the L76B is connected to the Raspberry Pi; the jumper cap is connected to C, and the Raspberry Pi is connected to the USB interface. Connection (the module can be used as a USB to TTL module at this time).

Question: Why is there no corresponding demo for CAT24C32 and what is its function?

Answer:

The role of CAT24C32 is to provide an ID EEPROM (including vendor information, GPIO mapping and valid device tree information) for Raspberry Pi, which is the Micro-HAT (uHAT) specification introduced by Raspberry Pi. Want to ensure consistency and compatibility with future add-on boards and provide a better user experience.

Question: Does the coin cell battery support charging?

Answer:

If a rechargeable button battery is used, the module can support charging the button battery on the back under normal power supply conditions. Rechargeable battery ML1220 is recommended。

Question: Why are the obtained GPS coordinates in Baidu Maps or Google Maps inaccurate?

Answer:

The raw GPS data obtained are unprocessed geographic coordinates, generally called lunar coordinates, which need to be added to a certain algorithm to be accurate. Different maps use different algorithms. The same coordinates are on Google and Baidu. processing is different. The program provides a Baidu map algorithm by default, which can be used as a reference.

Question: Why is the location printed by the Raspberry Pi 0, 0 or the direct program is stuck and cannot be located?

Answer:

Place it in the open air, with the antenna facing down, to ensure that there will be positioning data within 45 seconds after the device is turned on.

Question: What is the positioning error of the L76X GPS HAT in meters?

Answer:

Tested in an open environment with clear weather, the error is within 2.5m.

Question: How to parse the obtained positioning information?

Answer:

The following takes GNRMC as an example to parse the obtained positioning information, as shown below:

(/wiki/File:L76X_GPS_HAT_FAQ_006.png)

(/wiki/File:L76X_GPS_HAT_FAQ_007.png)

Question: How to change the baud rate of the module to 115200?

Answer:

Use the following commands to change it to "115200":

```
$PCAS01,5*19
```

Change it as the default "9600":

```
$PCAS01,1*1D
```

(/wiki/File:L76K_GPS_Module_FAQ.png)

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.
Working Time: 9 AM - 6 PM GMT+8 (Monday to Friday)

Submit Now (https://service.waveshare.com/)

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